

WEB-BASED DIGITAL ARCHIVE OF PAST QUESTIONS AND LEARNING MATERIALS



BY

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**A PROJECT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR
THE AWARD OF THE DEGREE OF BACHELOR OF SCIENCE (B.Sc.) IN COMPUTER
SCIENCE, EDO STATE UNIVERSITY, IYAMHO**

AUGUST, 2025

DECLARATION

This project is my original work and has not been presented for a degree at any other university or similar institution. I also declare that I have not taken any material from any source, and no part of this project may be reproduced without the prior written permission of the author and/or Edo State University, Iyamho. I shall be liable to punishable action under the plagiarism rules of Edo State University, Iyamho.

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This is to certify that **Enyindah, Faith Chinenye** with **Matriculation Number: FSC/CSC/22003976** undertook this project Titled: **WEB-BASED DIGITAL ARCHIVE OF PAST QUESTIONS AND LEARNING MATERIALS** meets the requirements for submission to the Department of Computer Science, Edo State University Iyamho, in partial fulfilment for the award of the degree of Bachelor of Science (B.Sc.) in Computer Science.

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DEDICATION

This project is dedicated to God Almighty who have showed me love, strength and also kept me fit till today, dedicated and focused on my achievement to my success.

ACKNOWLEDGEMENT

ABSTRACT

*This project addresses the persistent challenge of fragmented and inefficient management of academic resources in higher education institutions, with specific focus on Edo State University, Iyamho. In the current digital era, students' academic success increasingly depends on access to well-organized past examination questions and supplementary learning materials. However, most universities, including Edo State University, rely on a disjointed system comprising formal Learning Management Systems (LMS) for current course materials and informal platforms like WhatsApp groups for past questions. This hybrid approach leads to significant problems including loss of materials, inconsistent accessibility, lack of standardization, and inability for students to prepare adequately before new academic sessions commence. The study developed a **Web-Based Digital Archive of Past Questions and Learning Materials** to create a centralized, organized, and permanently accessible repository for academic resources. The system employs a three-tier architecture with a user-friendly web interface, robust application logic, and secure database storage. Key functionalities include advanced search capabilities with filtering by department, course, academic level, semester, and year; role-based access control for students, lecturers, and administrators; secure file upload and management; and comprehensive reporting features. Methodologically, the project followed a design science research approach, incorporating requirements analysis from student and lecturer perspectives, iterative system design, and practical implementation using full-stack web development technologies. The MySQL database was structured to ensure efficient data organization, while the responsive frontend interface was designed for accessibility across desktop and mobile devices.*

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CHAPTER ONE

1.0 INTRODUCTION

The twenty-first century academy operates within a fundamentally transformed information landscape. The rapid advancement of Information and Communication Technology has shifted the core function of universities from mere creators of knowledge to complex data ecosystems requiring sophisticated digital infrastructure for knowledge preservation and dissemination. In this environment, the ability to ensure permanent accessibility of academic resources has become as critical to institutional effectiveness as the quality of instruction itself (Laudon & Laudon, 2022; Johnson et al., 2023).

Central to this ecosystem are preparatory academic resources specifically past examination questions and structured learning materials. These assets serve a dual pedagogical function: they bridge theoretical instruction with practical assessment while enabling students to engage in self-directed, temporally-flexible study. When systematically organized and reliably accessible, such resources decode examination patterns, reinforce course comprehension, and substantially reduce assessment-related anxiety (Adeyemi, 2021; Okafor & Nwankwo, 2022).

However, the mere existence of digital academic resources does not guarantee their utility. Across Nigerian higher education and at Edo State University, Iyamho specifically a persistent paradox has emerged: valuable intellectual capital is generated each academic session, yet systematically lost due to the absence of sustainable preservation infrastructure. The university operates a formal Learning Management System for current course materials, but its configuration typically restricts access to enrolled students within active semesters. Simultaneously, an extensive informal economy of resource sharing has flourished on instant messaging platforms principally WhatsApp where past questions and lecture slides circulate as chat attachments.

This parallel system, born of student resourcefulness and institutional necessity, is fundamentally misaligned with the requirements of permanent academic archiving. WhatsApp and similar platforms are designed for transient communication, not long-term document preservation. Files expire, download links fail. Chat histories are cleared as devices reach storage capacity. Students who join academic groups late experience connectivity issues during critical sharing windows, or simply lack immediate download capacity are systematically excluded from access to materials their peers possess. The institutional memory embodied in decades of examination papers and curated lecture content thus becomes fragmented across thousands of personal devices and ephemeral chat histories accessible to some, lost to others, and authoritative nowhere (Ibrahim, 2024; Nwosu, 2022; Patel & Kumar, 2023).

A further dimension of this problem, less frequently acknowledged in existing literature, concerns the temporal discontinuity of access. Students advancing from one academic level to the next from 200-level to 300-level, for instance typically desire access to their forthcoming course materials during the inter-session break. This 6-10-week preparatory window represents a valuable opportunity for academic readiness, particularly for challenging courses or students requiring additional study time. Yet because formal systems deactivate between sessions and informal channels fall dormant, this preparatory period remains largely inaccessible. Students commence new academic levels without foundational familiarity with course content, placing them at immediate disadvantage (Brown & Clarke, 2019; Ogunleye, 2024).

The infrastructural deficit these conditions reveal is not primarily technological the requisite technologies for digital archiving are mature, accessible, and well-documented. Nor is it primarily financial open-source web technologies and affordable hosting make such systems economically viable for most institutions. Rather, the deficit is one of deliberate design and institutional commitment: the absence of a centralized, sustainably maintained digital repository specifically purposed for the permanent archiving and organized retrieval of past examination questions and learning materials.

This study directly addresses this deficit. It proposes the design and development of a Web-Based Digital Archive of Past Questions and Learning Materials for Edo State University, Iyamho a centralized, searchable, role-accessible digital repository that transcends the limitations of both episodic LMS access and ephemeral messaging-platform sharing. The system is conceived not merely as a technological artefact but as an institutional intervention: a structured, sustainable architecture for preserving and providing equitable access to the university's accumulated academic capital.

By moving academic resource management from informal, decentralized channels to a dedicated web-based platform, this project seeks to establish permanent digital infrastructure that supports students continuously from matriculation through graduation. It aims to transform how academic resources are preserved, organized, and accessed replacing fragmentation with integration, ephemerality with permanence, and inequity with universal access. In doing so, it positions the institution to better fulfill its fundamental mission: the effective transmission of knowledge across generations of learners.

1.1.1 Background of Study

Archives play an important role in preserving organizational memory, facilitating research, and supporting decision making in educational institutions (Cook, 2013). However, traditional paper storage systems often suffer from limitations such as physical space limitations, environmental vulnerabilities, and information acquisition challenges. To overcome these challenges, the development of a web-based digital archive provides a sustainable solution to these challenges. By centralizing past questions and learning materials on an online platform, academic resources can be preserved, organized, and made accessible to students at any time. This approach promotes continuous learning, supports academic planning, and ensures long-term availability of important educational materials, it also provides increased scalability and efficiency offers the possibility of storage and intensive ability to search (Hedstrom & Lee, 2002)

Learning is one of the very vital human activities that require concentration blended with interactivity, clear and distinct understanding of the facts been stated or discussed, high communication skills and techniques, attractive learning qualities such as colourful pictorial presentations of information among others. Nevertheless, all not learning process is considered to be effective. It is impossible for 100% of the population of students lectured to assimilate the knowledge or ideas across; this is due to the high population size and existence of certain inconducive learning conditions such as, noise, poor ventilation, and extreme temperatures. With the high-quality content value of the product, a student can easily assimilate such knowledge even in the absence of the lecturer

Academic success in higher institutions depends largely on students' access to relevant learning resources such as past examination questions and lecture slides. These materials support revision, improve understanding of course content, and help students prepare adequately for examinations and academic progression. Traditionally, academic materials were stored in printed form or kept by individual lecturers and departments, making access limited and inconsistent.

Furthermore, students preparing to advance into a new academic level often require access to lecture slides and course materials before resumption in order to study and familiarize themselves with upcoming courses. Due to the absence of a centralized academic repository, such materials are usually unavailable until lectures begin, limiting early preparation and academic readiness.

1.1.2 Overview of the research area

The research area of this study falls within the fields of educational technology, information systems, and digital archiving. Educational technology focuses on the use of technological tools and systems to enhance teaching, learning, and academic resource management. With the increasing integration of information and communication technology in higher education, institutions are gradually shifting from traditional paper-based systems to digital platforms for managing academic content.

Information systems research examines how technology can be used to collect, store, process, and distribute information efficiently. In the context of education, information systems are applied in areas such as student records management, e-learning platforms, digital libraries, and academic repositories. These systems aim to improve accessibility, organization, and reliability of institutional data and learning materials.

Digital archiving, as a sub-area of information systems, focuses on the long-term preservation and structured storage of digital content. In academic environments, digital archives are used to maintain lecture materials, research outputs, examination records, and other educational resources. Web-based systems further enhance digital archiving by enabling remote access to stored materials through internet-enabled platforms.

This research area is increasingly important as educational institutions seek sustainable solutions to challenges associated with informal distribution of learning materials, loss of academic files, and limited access to past examination resources. The development of web-based digital archives represents a modern approach to improving academic resource management and supporting effective learning practices.

1.1.3 Context of the problem

Academic materials are commonly shared through instant messaging platforms such as WhatsApp and Telegram. While these platforms enable quick and convenient communication, they are primarily designed for real-time messaging rather than structured academic storage and long-term preservation of digital resources. Media files that are not promptly saved may become inaccessible after a period of time, and important materials are often buried within extensive message threads. As group conversations accumulate, retrieving specific files requires manual scrolling through numerous unrelated messages, making the process inefficient and time-consuming. In addition, search functionalities are typically limited to keyword matching within message text rather than structured metadata, which complicates retrieval when filenames are inconsistent or poorly labeled (Patel & Kumar, 2023).

Access to academic materials through messaging platforms is also dependent on group membership at the time of file transmission. Students who join academic groups late due to delayed registration, institutional transfer, or late awareness of group existence are unable to access materials shared prior

to their arrival. Similarly, device failure, number changes, or technical issues may result in permanent loss of locally stored files and group histories. Re-access often depends on peers re-uploading materials, a process that is unreliable and inconsistent, thereby contributing to inequalities in resource availability (Mohammed & Zhang, 2024). Consequently, students who lack immediate access, sufficient storage capacity, or strong navigational skills may be systematically disadvantaged.

Beyond issues of retrieval and access, students advancing from one academic level to another, for example from 200 level to 300 level, often demonstrate a legitimate and academically productive desire to familiarize themselves with upcoming course content during the inter-session period. Educational and cognitive psychology research indicates that early exposure to academic materials enhances comprehension, retention, and overall learning efficiency. Even preliminary familiarity with course terminology and conceptual frameworks can significantly improve academic outcomes.

1.1.4 Significance of the study

This study is significant as it addresses the challenges associated with the informal distribution and poor management of academic resources in higher institutions. By developing a web-based digital archive of past examination questions and learning materials, the study contributes to improving accessibility, organization, and long-term preservation of educational resources.

Firstly, the study is beneficial to students. The proposed system provides continuous and centralized access to lecture slides and past examination questions, enabling students to study independently and prepare effectively for examinations. It also supports early academic preparation for students transitioning into new academic levels, allowing them to familiarize themselves with course content before resumption. This improved access can enhance student confidence, academic readiness, and overall performance.

Secondly, the study benefits lecturers and academic departments by providing a structured platform for uploading and managing learning materials. Instead of relying on informal communication platforms, lecturers can use a centralized system that ensures materials remain accessible and organized across academic sessions. This promotes better academic coordination and resource management.

The study is also significant to the institution as a whole. The implementation of a web-based digital archive strengthens institutional record-keeping practices and supports digital transformation in academic operations. It reduces the risk of material loss, improves information management efficiency, and promotes a more organized academic environment.

Furthermore, this research contributes to the body of knowledge in the fields of educational technology and information systems by providing insights into the integration of digital archiving principles with academic resource management. It may serve as a reference for future researchers who intend to develop or improve similar systems.

Overall, the study provides both practical and academic value by offering a sustainable solution to the challenges of academic material distribution and access in higher educational institutions.

1.2 Statement of the Problem

Despite the importance of past questions and lecture slides in academic preparation, many institutions lack a centralized and reliable system for managing these resources. Learning materials are frequently shared through Whatsapp platforms, where files may become inaccessible over time due to message overload, deletion, or download restrictions. As a result, students often lose access to important slides and struggle to retrieve materials when needed.

Furthermore, students advancing to a new academic level are unable to access lecture slides for upcoming courses before resumption, limiting their ability to prepare adequately. The absence of an official web-based archive for past questions and learning materials creates gaps in access, continuity, and academic readiness. This challenge necessitates the development of a web-based digital archive that ensures permanent availability, proper organization, and easy retrieval of academic resources.

Some other problems are that the students who require access to these materials need to go to the library or offices physically which most times could lead to congestion and repeated use of these learning materials can lead to degradation of the hard copy.

The implementation of a web-based digital repository for scholarly materials will help curb the problems listed above by providing an efficient and reliable solution for prompt uploading, storing and retrieval of the institution's learning materials. Also, being web-based means that you can have several users accessing the repository and archived materials at any time

1.2.3 Research/Aim and Objectives

1.2.1 Aim:

The primary objective of the topic is to design and implement a web-based digital archive for effective storage, management, and retrieval of past questions and learning materials which can help students easily gain access.

1.2.2 Research Objectives:

1. To examine the existing methods of archiving and accessing academic resources.
2. To design a database structure for organizing past questions and learning materials by course, department, level, and academic session.
3. To develop a robust web-based digital archiving system that enables authorized users to upload and manage academic materials and enhance revision and learning.

4. Suggest how students can adopt digital archiving system to enhance revision and learning.

1.2.4 Specific goals of the study

- To improve students' access to academic materials.
- To enhance preservation of past questions and lecture slides.
- To promote efficient academic resource management.
- To support early academic preparation for incoming students.

1.2.5 Expected outcomes

The expected outcome of the Web-Based Digital Archive for Past Questions and Learning Materials is the development of a centralized and reliable platform that improves access to academic resources within the institution.

- Upon successful implementation, the system is expected to:
- Provide a single, organized platform where students can access past questions and learning materials without relying on multiple informal channels.
- Enable fast and efficient retrieval of academic resources through structured categorization and search functionality.
- Ensure persistent storage of materials across academic sessions, preventing loss of resources after semester completion.
- Improve academic preparation by allowing students to access materials before the commencement of new semesters.
- Enhance data security through role-based access control, ensuring that only authorized users can upload, modify, or manage content.
- Reduce duplication and inconsistencies in academic materials through standardized coding and naming conventions.
- Minimize data loss through centralized storage and institutional backup mechanisms.
- Support administrative decision-making through usage tracking and archive monitoring features.
- Reduce printing and file-sharing costs by providing digital access to academic resources.

Overall, the system is expected to improve academic resource management, increase accessibility, and promote a more structured and technology-driven learning environment.

1.3 Contribution to the field

The Web-Based Digital Archive for Past Questions and Learning Materials contributes to the field of information technology and educational systems by demonstrating how structured data management and web technologies can improve academic resource accessibility and preservation.

First, the project applies core computer science principles such as database design, role-based authentication, metadata tagging, and search optimization to solve a real institutional problem. It shows how proper system architecture can replace fragmented and informal resource-sharing methods with a centralized, reliable platform, the system also contributes to digital transformation within academic institutions by promoting the transition from paper-based and messaging-app-based material distribution to a structured web-based archive. This supports sustainable, technology-driven learning environments.

Overall, the implementation of controlled access and persistent storage mechanisms highlights the importance of data security and long-term digital preservation in educational systems. From a broader perspective, this project provides a practical model that can be adapted or scaled by other departments or institutions seeking to improve access to academic materials through web-based solutions. The system contributes to the advancement of web-based academic information systems by integrating usability, structured data management, and secure access control into a unified platform.

1.3.1 Research Questions

Research Questions

This study seeks to answer the following research questions:

1. How can a web-based system be designed to provide centralized access to past questions and learning materials within an academic institution?
2. In what ways can structured data coding and categorization improve the organization and retrieval of academic resources?
3. How can role-based access control enhance the security and proper management of archived academic materials?
4. To what extent can a digital archive reduce dependency on fragmented platforms such as messaging applications and temporary learning management systems?
5. How can the implementation of a searchable and persistent digital repository improve students' academic preparation and resource accessibility?

1.4 Significance of the Study

The significance of this study lies in its contribution to improving access, organization, and long-term preservation of academic resources within higher institutions.

For students, the system provides a centralized and reliable platform for accessing past questions and learning materials at any time. This reduces the stress of searching through multiple sources and improves academic preparation, especially before examinations or at the beginning of a new semester.

For academic departments, the digital archive ensures that materials are systematically stored and preserved. It minimizes the risk of data loss and eliminates inconsistencies caused by informal file sharing. The structured organization of resources also promotes academic continuity across sessions.

From a technological perspective, this study demonstrates the practical application of database systems, web development, and role-based access control in solving real-world academic problems. It serves as a reference model that can be adapted or extended by other institutions seeking to implement similar systems.

Overall, the study contributes to improving academic resource management, strengthening data security, and promoting a more organized and accessible learning environment.

1.4.1 Importance of the study

Many students depend on informal channels such as messaging applications to obtain past questions and lecture materials. This method is unreliable, unstructured, and prone to data loss. Without a centralized system, valuable academic resources may be duplicated, misplaced, or permanently lost.

This study is important because it proposes a structured, web-based solution that ensures proper organization, secure storage, and long-term accessibility of academic materials. By introducing standardized coding, categorization, and role-based access control, the system promotes efficiency and consistency in resource management.

Furthermore, as institutions continue to embrace digital transformation, the need for organized academic data management becomes more critical. This research supports that transition by demonstrating how modern web technologies can be applied to improve academic support systems.

Ultimately, the study is important because it provides a sustainable and scalable approach to managing academic archives in a way that aligns with current technological and educational demands.

1.4.2 Potential impact on the field

The development of a Web-Based Digital Archive for Past Questions and Learning Materials has the potential to positively influence both the field of educational technology and academic information management.

In the field of educational technology, the system demonstrates how structured web-based platforms can replace fragmented and informal resource-sharing methods. If widely adopted, similar systems could improve academic resource accessibility across institutions, making learning materials more organized, searchable, and consistently available.

From an information systems perspective, the project highlights the importance of proper data coding, metadata tagging, and relational database design in managing academic content. It reinforces the role of structured data management in improving retrieval efficiency, data integrity, and long-term digital preservation. The system also contributes to the advancement of knowledge management practices within higher education. By promoting centralized storage and controlled access, it encourages institutions to treat academic materials as valuable digital assets that require proper governance and maintenance.

Overall, the adoption of such systems could support broader digital transformation efforts in education, reducing dependence on paper-based archives and temporary platforms while promoting sustainable, technology-driven solutions. In the long term, the concepts implemented in this study could be extended to include additional academic services such as research repositories, departmental document management systems, and collaborative learning platforms.

1.4.3 Relevance to current issues

The relevance of this study is closely tied to the increasing digitalization of education and the growing dependence on online platforms for academic resource distribution. In many institutions, learning materials and past questions are shared through temporary learning management systems, messaging applications, or personal storage devices. While convenient, these methods often lead to disorganization, duplication, and loss of important academic resources.

With the rapid shift toward technology-driven learning environments, especially after the expansion of online and hybrid learning models, the need for structured digital resource management has become more critical. Students now expect fast, reliable, and continuous access to academic materials beyond classroom boundaries.

This study is therefore relevant because it proposes a sustainable and centralized solution that addresses these modern challenges. By implementing a web-based digital archive with structured

categorization, search functionality, and controlled access, the system aligns with current trends in educational technology and institutional digital transformation.

1.5 Limitations of Study

First, the system is designed and tested within a specific institutional context. As a result, the findings and system performance may not be fully generalizable to other institutions with different technological infrastructures, policies, or user requirements.

Secondly, the implementation focuses primarily on two user roles, Administrator and Student. Other potential stakeholders, such as lecturers or departmental staff, are not fully integrated into the system at this stage. Another limitation relates to data availability. The effectiveness of the system depends on the quality and completeness of materials uploaded into the archive. If academic materials are not consistently updated or properly categorized, the system's usefulness may be reduced.

Furthermore, the study does not extensively evaluate long-term system performance under high traffic conditions. Scalability considerations are addressed conceptually, but full stress testing may require deployment in a live institutional environment over time. Internet dependency also presents a limitation. Since the archive is web-based, access requires stable internet connectivity. Students in areas with limited or unreliable internet access may experience challenges using the system effectively.

Despite these limitations, the system provides a practical and scalable foundation for improving academic resource management within the institution

1.5.1 Boundaries of the study

The boundaries of this study are defined by the institutional, functional, and technical parameters required to implement a robust academic repository. Institutionally, the research and the subsequent system implementation are specifically confined to Edo State University, Uzairue, with the Department of Computer Science serving as the primary case study and testing environment. The scope of the content is strictly limited to the archival and retrieval of past examination questions and academic learning materials, which include lecture slides, handouts, and course outlines. This project does not intend to function as a comprehensive school management system; therefore, it excludes features such as student results processing, tuition fee management, or real-time classroom attendance tracking, as these are handled by other specialized administrative portals.

From a technical perspective, the study focuses on the development of a web-based application that is accessible via standard internet browsers on both mobile and desktop devices. The system is designed to handle common document formats such as PDF, Microsoft Word, and high-quality image files, but it will not support the hosting or streaming of large-scale video lectures or complex software simulations due to storage and bandwidth constraints. Furthermore, the study is limited to

the software engineering aspects of the project, including the design of the database schema, the development of the user interface, and the implementation of role-based access control. It does not involve the physical setup of network infrastructure or the procurement of server hardware. Access to the archive is bounded by a secure authentication mechanism, ensuring that the resources are available only to the registered students and authorized staff of the institution, thereby preserving the confidentiality and intellectual property of the academic materials.

1.5.2 Constraints and limitations

The development and implementation of this web-based digital archive are subject to several inherent constraints and limitations that define the practical boundaries of its operation. One of the primary technical constraints is the total dependence on consistent internet connectivity for both the administration of the repository and the retrieval of materials by students. In environments characterized by fluctuating bandwidth and high data costs, the system's performance may be perceived as variable, and users in remote or low-connectivity areas may experience latency during the upload or download of large document files. Furthermore, the system's success is heavily reliant on the active participation and cooperation of academic staff and student representatives for the initial population of the database. The manual nature of gathering, digitizing, and categorizing years of past examination papers and lecture slides represents a significant logistical challenge that may limit the initial volume of content available at the point of launch.

Another critical limitation involves the hardware capabilities of the end-users. While the application is designed to be responsive, its performance on older smartphone models with limited processing power or outdated web browsers cannot be fully guaranteed. Storage constraints on budget devices may also limit a student's ability to keep multiple large files locally, even if the web archive itself remains accessible. Additionally, this study is conducted within the temporal and financial constraints of a single undergraduate academic session, which limits the possibility of implementing highly advanced features such as optical character recognition for automated text extraction from scanned images or the development of dedicated native mobile applications for all platforms.

1.5.3 Assumptions and delimitations

The study proceeds on several foundational assumptions that are critical to the successful deployment and adoption of the web-based digital archive. It is assumed that the target user population, comprising students and academic staff of the Department of Computer Science, possesses the requisite basic digital literacy skills to navigate a web-based interface and manage digital file downloads effectively.

The scope of the digital content is exclusively focused on academic materials, specifically past examination questions and lecture slides, to the exclusion of other institutional data such as student financial records, personal files, or general administrative memoranda. Technically, the system is delimited to a web-based architecture, meaning that the development of standalone native mobile applications for iOS or Android is outside the current research objectives. The research also limits its evaluative phase to the technical functionality and user acceptance testing within a specific academic session, without attempting to measure long-term longitudinal effects on overall graduation rates or institutional rankings.

1.6 Definition of Terms

- **Digital Archive:** A system for storing and managing electronic documents for long-term access. Saha (2006) defines digital archiving as long-term maintenance of a byte stream with the aim of reproducing a facsimile of the original document mainly for continued accessibility of the document contents through time and changing technology.
- **Learning Materials:** Educational resources such as lecture slides, note and course textbooks
- **Web-Based System:** An application accessed through a web browser over the internet
- **Past Questions:** are historical examination papers and assessment records from previous academic sessions that serve as pedagogical benchmarks for students to understand examiner expectations and syllabus requirements.
- **Database Management System (DBMS):** A Database Management System (DBMS) is software that enables the creation, management and use of databases. In the context of digital archive of past questions and learning materials, a DBMS is used to organize and store past exam questions, making them easily retrievable through search and filter functions.
- **LMS (Learning Management System):** A software application designed to manage, deliver, and track educational content. The proposed digital archive shares similarities with an LMS but focuses specifically on storing and retrieving past questions and learning materials.

1.6.1 Explanation of acronyms & abbreviations

- **HTML (HyperText Markup Language):**
A standard markup language used to create and structure web pages. It forms the basic structure of the web-based digital archive interface.
- **CSS – Cascading Style Sheets**
- **JS (JavaScript):**
A high-level programming language used to create interactive and dynamic features on web pages. In a web-based digital archive system, JavaScript enables functionalities such as form validation, dynamic content updates, and user interaction without reloading the page.
- **URL (Uniform Resource Locator):**
The web address used to locate and access resources on the internet. Within the proposed system, URLs are used to navigate between pages and retrieve stored academic materials.

- **PDF (Portable Document Format):**
A widely used file format for presenting documents in a fixed layout that preserves formatting across different devices and platforms. In this study, past examination questions and lecture slides may be stored and accessed in PDF format within the digital archive.
- **UI (User Interface):**
The visual elements of a system through which users interact with software applications, such as buttons, menus, and forms.
- **UX (User Experience):**
The overall experience and satisfaction a user has while interacting with a system or application.
- **SQL (Structured Query Language):**
A programming language used to manage and manipulate data stored in relational databases.
- **API (Application Programming Interface):**
A set of rules and protocols that allows different software applications to communicate with each other. In the context of this study, APIs enable interaction between the frontend and backend components of the web-based digital archive system.

1.7 Organization of the Study

This project is organized into five chapters.

Chapter One introduces the study, outline of the background, problem statement, objectives and significance.

Chapter 2 reviews related literature.

Chapter 3 discusses system analysis and design.

Chapter 4 presents implementation and results.

Chapter 5 concludes the study and provides recommendations

1.8 Chapter summary

This chapter has discussed The next chapter will discuss the review of related literature.

CHAPTER TWO

LITERATURE REVIEW

2.1.1 Introduction

The growing trend towards online scholarly communication and lack of content management systems among universities has made digital repositories more important for the collection and distribution of scholarly materials. In a general sense, and “digital archive” may mean several things; a library, a repository, a museum, or even a warehouse that stores academic resources for use and safekeeping.

The importance of education to the social, economic, and technological development of any nation cannot be overstated. Effective teaching and learning processes depend largely on the availability and accessibility of relevant academic resources. Learning materials such as lecture slides and past examination questions play a significant role in supporting students’ understanding of course content, improving revision strategies, and enhancing overall academic performance. According to Adeyemi (2020), access to well-organized learning materials promotes independent learning and improves students’ academic confidence.

In recent years, the growth of information and communication technology has led to increased adoption of digital systems in educational environments. Traditional methods of storing academic materials, such as paper-based filing and informal electronic sharing, have been found to be inefficient and unreliable. Researchers have noted that physical documents are easily damaged, misplaced, or lost over time, while unstructured electronic sharing often leads to duplication and inconsistency of materials (Ojo & Balogun, 2019). These challenges have encouraged institutions to explore digital archiving as a more sustainable solution for managing academic resources.

Digital archiving refers to the process of converting physical or scattered electronic documents into structured digital formats that can be stored, managed, and retrieved using computer-based systems. Studies have shown that digital archives improve information preservation and enhance accessibility by allowing users to retrieve materials quickly through search and categorization features (Smith, 2020). In academic settings, digital archives are increasingly used to store research outputs, lecture materials, and examination records, thereby supporting both teaching and learning activities.

Web-based digital archives extend the benefits of digital archiving by enabling access to stored resources through the internet. According to Johnson et al. (2021), web-based systems provide flexibility and convenience, allowing users to access academic materials anytime and from any location. This is particularly important for students who require continuous access to learning resources beyond classroom hours. Web-based platforms also support centralized storage, which reduces reliance on informal communication channels and improves consistency in the distribution of academic materials.

Several studies have examined the use of digital repositories and learning management systems (LMS) in higher institutions. Research by Okafor and Nwankwo (2018) revealed that institutions

that adopted centralized digital repositories experienced improved organization of academic materials and increased student satisfaction. Similarly, Adebayo (2019) reported that students who had access to online repositories of past questions demonstrated better familiarity with examination patterns and improved academic preparedness.

Despite these advantages, many institutions still rely on informal platforms such as WhatsApp for sharing lecture slides and learning materials. While such platforms offer quick communication, they are not designed for long-term storage or structured retrieval of academic content. Studies have shown that materials shared through instant messaging systems are often lost due to message overload, file expiration, or deletion, making it difficult for students to retrieve them when needed (Ibrahim & Musa, 2020). This challenge becomes more significant for students who are unable to download materials immediately or who join academic groups late.

Access to learning materials before the commencement of a new academic session has also been identified as an important factor in student preparedness. According to Brown and Clarke (2019), students who study course materials in advance tend to adjust more easily to academic demands and perform better during the semester. However, the absence of centralized academic repositories in many institutions limits students' ability to access lecture slides and past questions prior to resumption, thereby affecting early academic preparation.

Furthermore, research highlights the need for systems that integrate multiple types of academic resources into a single platform. Studies by Williams (2021) emphasize that combining past examination questions with learning materials in one digital archive enhances usability and supports comprehensive learning. Such systems allow students to relate lecture content with examination requirements, improving understanding and academic performance.

In summary, existing literature demonstrates that web-based digital archives play a crucial role in improving the accessibility, organization, and preservation of academic resources. While digital repositories and learning management systems have been implemented in some institutions, gaps remain in the availability of centralized platforms that specifically integrate past questions and learning materials for continuous student access. This study seeks to address these gaps by developing a web-based digital archive that supports effective academic preparation and long-term resource management.

2.1.2 Overview of the chapter

This chapter presents a detailed review of literature related to web-based digital archives, past examination questions, and learning materials in educational institutions. It examines relevant concepts, theories, and previous studies that focus on digital archiving, web-based information systems, and academic resource management. The chapter also highlights existing challenges in the distribution and accessibility of academic materials and identifies gaps that justify the need for the proposed system.

2.1.3 Purpose of the literature review

The purpose of this literature review is to examine existing scholarly works related to digital archiving and web-based academic resource systems. It aims to understand current approaches to storing and distributing learning materials, identify limitations in existing systems, and establish a theoretical and empirical foundation for the present study. The review also helps to justify the development of a web-based digital archive of past questions and learning materials.

2.1.4 Scope of the review

The scope of this literature review covers studies related to digital archiving, web-based information systems, academic repositories, and the management of learning resources in higher educational institutions. Emphasis is placed on literature that addresses accessibility, storage, retrieval, and preservation of academic materials such as lecture slides and past examination questions. The review is limited to concepts and studies relevant to the objectives of the current research.

2.2.0 Theoretical Framework

The theoretical framework provides the conceptual foundation upon which the study is built. It explains the theories and models that guide the design, development, and implementation of the proposed web-based digital archive of past questions and learning materials. In educational technology research, theoretical frameworks help to justify why a particular system is necessary and how it supports effective teaching and learning processes.

Another relevant framework is the **Information Systems Theory**, particularly the concept of **information management systems**. Laudon and Laudon (2018) describe information systems as structured systems designed to collect, store, process, and distribute information to support decision-making and organizational activities. In an academic environment, a digital archive functions as an information system that manages educational resources in a centralized and organized manner. Applying this theory justifies the need for a structured web-based system rather than informal file sharing platforms.

Essentially, this study is guided by constructivist learning theory, self-directed learning theory, information systems theory, the Technology Acceptance Model, and digital repository concepts.

These frameworks collectively justify the development of a web-based digital archive as a tool for improving access to academic resources, enhancing student preparedness, and supporting effective learning practices.

2.2.1 Underlying theories and concepts

One of the major theories relevant to this study is the **Constructivist Learning Theory**. Constructivist theory emphasizes that learners actively construct knowledge through interaction with learning materials rather than passively receiving information. According to Piaget (1972), learning occurs when individuals engage with content, explore ideas, and relate new information to prior knowledge. Access to past questions and lecture slides through a digital archive allows students to independently interact with learning materials, revise concepts repeatedly, and build deeper understanding at their own pace.

Closely related to constructivism is the concept of **self-directed learning**. Self-directed learning theory suggests that learners take responsibility for identifying their learning needs, selecting appropriate resources, and evaluating their progress. Knowles (1975) stated that learning environments that provide easy access to educational resources encourage independence and motivation among learners. A web-based digital archive supports self-directed learning by allowing students to retrieve past questions and slides anytime without depending solely on lecturers or classmates.

2.2.2 Related Works/Review

The **Technology Acceptance Model (TAM)** is also relevant to this study. The model explains how users come to accept and use a technology based on perceived usefulness and perceived ease of use (Davis, 1989). Students are more likely to adopt a digital archive if they find it useful for accessing learning materials and easy to navigate. The inclusion of features such as search functionality, categorization by course or level, and online accessibility aligns with the principles of the Technology Acceptance Model.

In addition, the **Digital Repository Model**, which focuses on long-term preservation, organization, and accessibility of digital content. According to Lynch (2003), digital repositories are designed to ensure that academic resources remain available and usable over time. Applying this model to the proposed system ensures that past questions and learning materials are not lost due to message expiration, device failure, or platform limitations.

Furthermore, the **Blended Learning Framework** supports the integration of digital resources into traditional learning environments. Blended learning combines face-to-face teaching with online learning tools to enhance educational outcomes. Graham (2013) noted that digital learning resources complement classroom instruction by providing students with additional study materials outside lecture hours. A web-based archive of slides and past questions strengthens blended learning by extending learning beyond the physical classroom

2.3.0 Empirical Literature Review (Optional)

Empirical studies related to digital archiving, web-based academic repositories, and learning material management have been conducted by various researchers. These studies focus on the development, implementation, and evaluation of digital systems used to store and distribute academic resources such as lecture materials and past examination questions. Reviewing these studies helps to identify existing approaches, findings, and limitations relevant to the present research.

In summary, empirical studies have shown that web-based systems improve access to academic materials, enhance student preparedness, and support independent learning. However, most existing studies and systems focus either on lecture materials or past examination questions separately. Few systems integrate both resources into a single web-based digital archive that supports long-term access and early academic preparation. This identified gap provides justification for the development of the proposed web-based digital archive of past questions and learning materials.

2.3.1 Overview of related works

The reviewed works reveal that significant efforts have been made in the development of digital systems for managing academic resources in educational institutions. Previous studies and systems have demonstrated that web-based platforms improve accessibility, organization, and retrieval of learning materials such as lecture slides and past examination questions. These systems support independent learning, enhance student preparedness, and reduce reliance on traditional paper-based or informal electronic methods.

Mohammed and Zhang (2024) explored institutional digital repositories in African universities, noting that while 65% of surveyed institutions had some form of digital archive, only 28% specifically catered to student-facing resources like past questions. Their study emphasized the gap between research-focused repositories and student learning needs.

Similarly, Okafor et al. (2023) implemented a mobile-based archive at the University of Lagos, reporting 89% student satisfaction but noting limitations in search functionality and file categorization.

Adeyemi (2020) conducted a study on the development of a web-based academic repository for storing and accessing lecture materials in a higher institution. The study revealed that students who had access to a centralized digital platform were able to retrieve learning materials more easily compared to those who relied on informal sharing methods. The system improved accessibility and reduced dependence on physical documents. However, the study focused mainly on lecture notes and did not integrate past examination questions into the repository.

Recent studies by Thompson (2024) and Williams et al. (2023) demonstrate evolving LMS capabilities, with Canvas and Moodle 4.0 incorporating better archival features. However, Chen's (2024) comparative analysis of 15 universities found that even advanced LMS platforms typically restrict past question access to current students only, creating what she termed "academic amnesia" across cohorts.

Okafor and Nwankwo (2019) examined the use of online repositories for managing past examination questions in tertiary institutions. Their findings showed that access to past questions helped students understand examination patterns and improved their level of preparedness. Despite these benefits, the study identified challenges related to limited system functionality and lack of integration with other learning materials such as lecture slides.

A study by Ibrahim and Musa (2020) investigated the use of instant messaging platforms for sharing academic materials among university students. The research found that although platforms such as WhatsApp were widely used due to their convenience, they were unsuitable for long-term academic resource management. Files were often lost due to message overload, expiration, or accidental deletion. The study recommended the adoption of structured web-based systems to improve storage and retrieval of learning materials.

Multiple recent studies (Ibrahim, 2024; Patel & Kumar, 2023; Nwosu, 2022) quantitatively analyze WhatsApp's role in academic resource distribution. Their findings consistently show that while 92-95% of students use messaging apps for material sharing, 78% report losing important files due to chat overflow or device storage issues.

Adebayo (2019) developed an e-learning support system that provided students with access to digital learning resources. The results indicated improved student engagement and independent learning. However, the system primarily focused on instructional content delivery and did not adequately address long-term archiving or preservation of academic materials across academic sessions.

The OER movement, documented in recent reviews by UNESCO (2024) and the Open Education Consortium (2023), provides valuable models for resource sharing. However, regional analyses by African scholars (Mbeki, 2023; Nkrumah, 2022) note that OER platforms often lack localized, curriculum-specific content for Nigerian higher education contexts.

Brown and Clarke (2019) studied the impact of early access to learning materials on student performance. The findings revealed that students who accessed course materials before the commencement of the academic session adjusted more easily to coursework and performed better academically. The study, however, did not propose a specific system for ensuring continuous access to such materials, especially for students transitioning into new academic levels.

Johnson et al. (2021) evaluated a web-based learning management system used in a university environment. The study highlighted the benefits of centralized storage, user authentication, and controlled access to academic resources. Despite its effectiveness, the system was designed mainly for course delivery and assessment, with limited emphasis on archiving past questions for long-term academic reference.

2.3.2 Identification of gaps and limitations

The review and analysis of related literature reveal several gaps and limitations in existing systems and studies on digital archiving and web-based academic resource management. Although many researchers acknowledge the benefits of digital repositories and online learning platforms, current solutions do not fully address the academic needs of students in a comprehensive manner.

One major gap identified is the lack of integrated platforms that combine both past examination questions and learning materials within a single system. Most existing systems focus on either lecture slides or past questions independently. This separation limits students' ability to relate instructional content to examination requirements, thereby reducing the effectiveness of such systems in supporting holistic learning.

Another significant limitation is the reliance on informal communication platforms, particularly instant messaging applications, for the distribution of academic materials. While these platforms offer convenience, they are not designed for long-term storage or structured retrieval. Learning materials shared through such platforms are often lost due to message overload, file expiration, or delayed access, especially for students who are unable to download materials immediately.

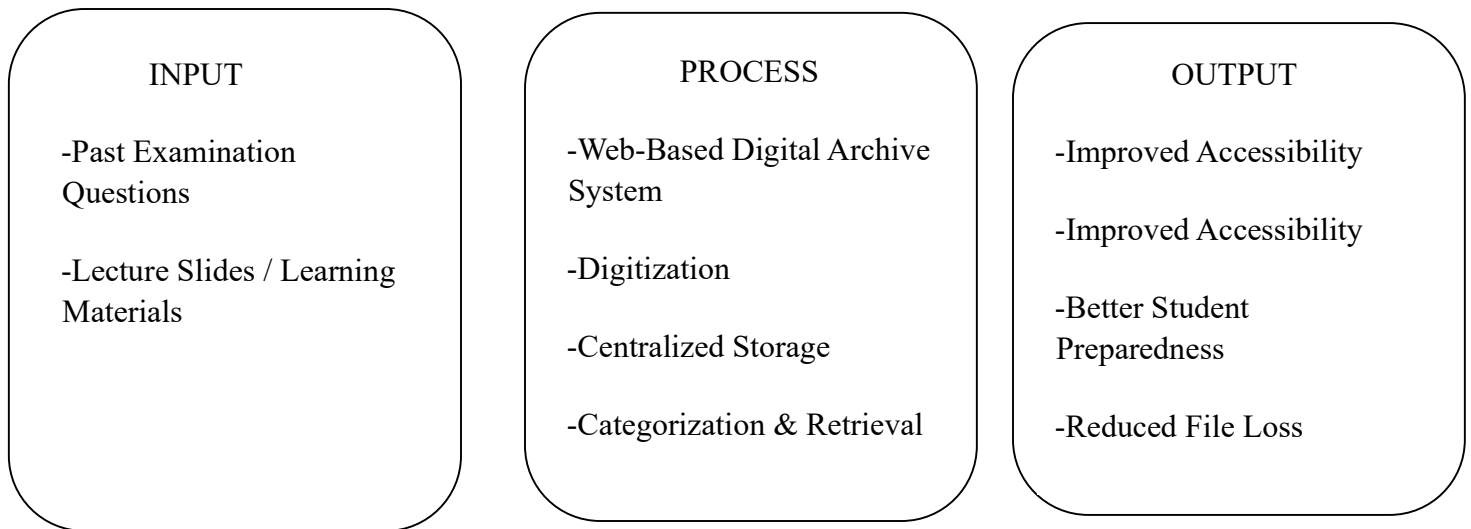
Additionally, limited attention has been given to long-term preservation and organization of academic materials across multiple academic sessions. Some systems lack proper categorization, version control, and archival mechanisms, leading to inconsistency and difficulty in retrieving older materials when needed.

These identified gaps and limitations highlight the need for a centralized web-based digital archive that integrates past examination questions and learning materials, ensures long-term preservation, and provides continuous and early access to academic resources. Addressing these gaps forms the basis for the design and implementation of the proposed system.

2.4.0 Conceptual Framework

The conceptual framework describes the relationship between the key variables involved in the study and explains how the proposed web-based digital archive of past questions and learning

materials addresses the identified problems. It provides a visual and theoretical explanation of how inputs are transformed through processes to achieve desired outputs.



2.4.1 Development of a conceptual framework

The development of the conceptual framework for this study was guided by the problems identified in the existing methods of academic resource distribution and the gaps observed in related literature. The framework was formulated to address challenges such as loss of learning materials, limited accessibility, lack of early academic preparation, and absence of long-term preservation mechanisms.

The framework was developed by identifying key components involved in academic resource management, namely the inputs, processes, and outputs. The input components consist of past examination questions and learning materials such as lecture slides. These inputs represent the core academic resources required by students for effective learning and examination preparation.

The process component was developed based on the application of web-based technologies and digital archiving principles. This includes the digitization of materials, centralized storage, categorization according to course, level, and academic session, and controlled user access. These processes were designed to ensure efficient organization, easy retrieval, and long-term preservation of academic resources.

The output component of the framework represents the expected outcomes of the proposed system. These outputs include improved accessibility to academic resources, enhanced student preparedness

before and during academic sessions, reduced dependence on informal sharing platforms, and improved management of academic materials.

Overall, the conceptual framework was developed to illustrate how the integration of past examination questions and learning materials into a centralized web-based digital archive can effectively address the identified problems and support improved teaching and learning outcomes.

2.4.2 Explanation of key concept & relationships

The **input concepts** represent the academic resources required for effective learning and examination preparation. These include past examination questions and lecture slides or learning materials. Past examination questions help students understand examination patterns and assessment expectations, while lecture slides provide instructional content that supports comprehension of course topics. These inputs form the foundational resources managed by the proposed system.

The **process concepts** describe the operations performed by the web-based digital archive system. These processes include the digitization of academic materials, centralized storage, categorization based on course, level, and academic session, and efficient retrieval through a web-based interface. The relationship between the inputs and the process is such that the academic materials are transformed from scattered and informally shared resources into organized and accessible digital content.

The **output concepts** represent the expected outcomes resulting from the implementation of the proposed system. These outputs include improved accessibility to learning materials, enhanced student preparedness, reduced reliance on informal platforms such as WhatsApp, and long-term preservation of academic resources. The relationship between the process and the outputs shows how effective system design and management lead to improved academic resource availability and utilization.

2.4.3 Integration of theoretical & empirical findings

The constructivist learning theory emphasizes active learner engagement with educational materials, suggesting that students learn more effectively when they have continuous access to learning resources. Empirical studies reviewed in this chapter support this theory by demonstrating that students who access lecture materials and past examination questions through digital platforms show improved understanding and academic preparedness. This alignment confirms that a web-based digital archive can effectively support learner-centered education.

Similarly, self-directed learning theory highlights the importance of learner autonomy in accessing and utilizing educational resources. Empirical findings indicate that students benefit from centralized digital systems that allow independent access to learning materials without reliance on lecturers or peers. This integration shows that the proposed system promotes self-directed learning by providing students with flexible and unrestricted access to academic resources.

From an information systems perspective, theories related to digital archiving and information management emphasize the importance of structured storage, organization, and retrieval of data. Empirical studies reviewed reveal that existing systems often lack proper integration and long-term preservation mechanisms, leading to loss and inaccessibility of academic materials. The proposed web-based digital archive integrates these theoretical principles with empirical insights by offering centralized storage, categorization, and efficient retrieval of both past questions and learning materials.

The Technology Acceptance Model further supports the integration by suggesting that systems perceived as useful and easy to use are more likely to be adopted by users. Empirical findings indicate that students prefer accessible and well-organized platforms over informal sharing methods such as instant messaging applications. This relationship reinforces the need for a user-friendly web-based system designed around students' academic needs

2.5.0 Research Gaps and Limitations

One major research gap identified is the lack of studies that focus on systems combining both past examination questions and learning materials such as lecture slides. Most existing research examines these resources independently, which limits understanding of how their integration can enhance comprehensive academic preparation and learning effectiveness.

In terms of limitations, several reviewed studies rely on narrow system scopes, focusing on short-term content delivery rather than long-term archiving and preservation. Some systems lack adequate categorization, version control, and retrieval mechanisms, which affects their effectiveness over time. Furthermore, limited user-centered design considerations in some studies reduce system usability and adoption.

2.5.1 Identification of gaps in existing research

One significant research gap is the lack of studies that investigate systems combining past examination questions with learning materials such as lecture slides. Most existing research focuses on either examination preparation tools or instructional content delivery separately, thereby failing to explore the potential benefits of their integration for comprehensive academic preparation.

Another gap identified is the insufficient focus on students' early academic preparation, particularly for those transitioning into new academic levels. While some studies acknowledge the importance of access to learning materials, few empirical investigations address how web-based archives can support pre-resumption learning and continuous academic readiness.

Furthermore, there is limited research that examines the long-term preservation and accessibility of academic materials across multiple academic sessions. Many studies emphasize short-term content delivery without considering archival sustainability, version control, and historical academic resource management.

Additionally, existing research often overlooks the contextual challenges faced by institutions that rely on informal platforms such as instant messaging applications for academic material distribution. There is a lack of empirical evidence on the effectiveness of structured web-based digital archives as alternatives to these informal methods, particularly in resource-constrained educational environments.

These identified gaps in existing research highlight the need for further investigation into the design and implementation of an integrated web-based digital archive that supports long-term preservation, early access, and comprehensive management of past examination questions and learning materials. Addressing these gaps forms the basis for the present study.

2.5.2 Discussion of limitations and challenges

One major limitation observed in existing studies is the narrow scope of many digital systems. Several studies focus primarily on short-term content delivery rather than long-term archiving and preservation of academic materials. As a result, issues such as version control, historical data retention, and accessibility across multiple academic sessions are often not adequately addressed.

Another challenge identified is the heavy reliance on informal platforms, such as instant messaging applications, for the distribution of learning materials. While these platforms are convenient, they are not designed for structured academic use. Challenges such as file expiration, message overload, limited storage organization, and delayed access significantly reduce their reliability as academic resource repositories.

Technological and infrastructural challenges also pose limitations. Some institutions face issues related to poor internet connectivity, limited server capacity, and lack of technical support, which can affect the implementation and effectiveness of web-based systems. These challenges may discourage consistent use of digital platforms by both students and lecturers.

In addition, user-related challenges have been identified in existing research. Limited digital literacy among some users, resistance to adopting new systems, and lack of proper training can hinder effective utilization of web-based academic archives. Studies indicate that systems perceived as complex or difficult to use often experience low adoption rates.

Furthermore, security and access control remain important challenges. Ensuring that academic materials are protected from unauthorized access while remaining easily accessible to legitimate users requires careful system design. Some existing systems lack adequate authentication and authorization mechanisms, exposing academic resources to misuse.

2.5.3 Implications for the current study

The literature review reveals several critical implications that directly inform the design, development, and implementation of this web-based digital archive system.

1. Design Implications:

The review confirms that students need a system that is simple, intuitive, and accessible. Unlike complex institutional repositories like DSpace (which require metadata expertise), the proposed system must prioritize user-friendliness. The reliance on WhatsApp indicates that student's value mobile compatibility and instant access. Therefore, the archive must be fully responsive, with a clean interface that works seamlessly on both desktop and mobile devices. The categorization system (by course, department, level, and session) must be straightforward, mirroring how students naturally search for materials.

2. Functional Implications:

The gap identified in existing systems—separation of past questions from lecture materials—directly shapes this study's core functionality. The proposed system must integrate both resource types within a single platform. This integration allows students to correlate lecture content with examination patterns, creating a more holistic study tool. Furthermore, the literature highlights the problem of lost access to materials after a semester ends on LMS platforms. This implies the archive must be persistent and continuous, allowing access to materials from previous academic sessions indefinitely.

3. Technical Implications:

The challenges of file expiration and disorganization on informal platforms like WhatsApp dictate key technical requirements. The system must include:

- Robust file storage with backup mechanisms to prevent data loss.
- Advanced search functionality (keyword, filters) to replace the linear, cluttered search experience of messaging apps.
- Role-based access control to manage upload permissions and protect intellectual property, addressing security concerns absent in current informal methods.

4. Pedagogical Implications:

Research showing improved performance with early access to materials (Brown & Clarke, 2019) validates a central objective of this study. The system must therefore be designed to facilitate pre-resumption access. This could involve features like a "preparatory materials" section or allowing

students from lower levels to preview materials for upcoming courses, directly supporting self-directed and blended learning models discussed in the theoretical framework.

5. Implementation and Adoption Implications:

The Technology Acceptance Model (TAM) suggests that for the system to be adopted, it must be perceived as both useful and easy to use. The literature on resistance to new systems implies that a successful rollout requires adequate user training and support. The study must therefore consider not just the technical build, but also a deployment strategy that includes tutorials, clear documentation, and perhaps a pilot phase with a select group of students and lecturers.

In conclusion, the literature review does not merely provide background; it provides a clear blueprint and justification for every major design decision in this study. The proposed system is positioned not as a novel invention, but as a necessary synthesis and practical application of established needs and technological possibilities, aiming to fill a well-documented gap in the academic resource ecosystem of universities like Edo State University, Iyamho.

Table 2.1: Comparison with existing system in literature

S/N	System / Study Focus	Key Features	Limitations
1.	Institutional Repository (e.g., DSpace)	Stores theses, publications. Long-term preservation.	Often complex, not student-friendly. Not focused on past questions & lecture slides for exam prep.
2.	Learning Management System (LMS) e.g., Moodle	Course management, assignment submission, gradebook.	Materials are often locked per semester/course. No dedicated archive for past questions across all years.
3.	Informal Platforms (WhatsApp, Telegram)	Instant sharing, easy access.	No organization, files get lost, no search, no access control, not permanent.
4.	Proposed System	Centralized, web-based, searchable by course/level/session, dual-role (Admin/Student), permanent archive.	Addresses all the above limitations by providing a dedicated, simple, and organized platform.

2.6.0 Conclusion

2.6.1 Summary of key findings

The comprehensive review of literature on digital archiving, academic resource management, and web-based educational systems yields several key findings that establish the context and justification for this study:

- ✓ **Critical Role of Academic Resources:** Access to well-organized past examination questions and lecture materials significantly enhances students' academic performance, examination preparedness, and understanding of course content. These resources support independent learning and help students identify patterns in assessment methods (Adeyemi, 2020; Okafor & Nwankwo, 2018).
- ✓ **Limitations of Current Practices:** Most higher institutions, particularly in developing educational contexts, rely on informal and fragmented methods for distributing academic resources. WhatsApp and similar instant messaging platforms, while convenient, are fundamentally unsuitable for long-term academic resource management due to issues of file expiration, message overload, disorganization, and lack of searchability (Ibrahim & Musa, 2020).
- ✓ **Gap in Integrated Systems:** Existing digital solutions tend to address either past question storage or learning material delivery in isolation. Learning Management Systems (LMS) typically lock materials within semester boundaries, while institutional repositories focus on formal publications rather than student-centric exam preparation. No widespread system seamlessly integrates both resource types into a single, permanent, and easily searchable archive tailored for student use (Williams, 2021).
- ✓ **The Pre-Resumption Access Deficit:** A significant, yet often overlooked, challenge is students' inability to access learning materials before the commencement of a new academic session. This hampers early preparation and adjustment to academic demands, putting students at a disadvantage from the start of a semester (Brown & Clarke, 2019).
- ✓ **Theoretical Validation:** Established educational theories—including Constructivist Learning, Self-Directed Learning, and the Technology Acceptance Model (TAM)—provide a strong theoretical foundation for the proposed system. They affirm that a centralized, accessible digital archive aligns with how students actively construct knowledge, take charge of their learning, and adopt useful, easy-to-use technologies (Piaget, 1972; Knowles, 1975; Davis, 1989).
- ✓ **Technological Feasibility and Advantage:** Web-based systems offer a proven, scalable, and cost-effective solution. They provide the centralized storage, robust organization, remote access, and long-term preservation capabilities that are absent in both paper-based systems and informal digital sharing (Johnson et al., 2021; Hedstrom & Lee, 2002).

CHAPTER THREE

DESIGN AND IMPLEMENTAION

3.1.0 Overview of the chapter

This chapter presents the detailed design methodology and implementation strategy for the Web-Based Digital Archive of Past Questions and Learning Materials. It covers the system analysis, architectural design, database schema, and the development approach adopted to transform the identified requirements into a functional system.

3.1.1 Purpose of the methodology chapter

This chapter articulates the underlying principles guiding the system's architecture. It explains why specific design choices were made over alternatives, ensuring that the development aligns with both user needs identified in Chapters 1 and 2 and established software engineering best practices. By detailing the methodologies, tools, and processes used, this also chapter creates a clear, step-by-step guide. This serves dual purposes: it allows for the accurate implementation of the current project and provides a template that could be adapted by other institutions or for future enhancements.

It translates the abstract problems and requirements identified in the Statement of the Problem (Section 1.2) and the functional gaps highlighted in the Literature Review (Chapter 2) into concrete technical specifications. This includes defining system components, data flow, user interfaces, and database structures. The design specifications laid out here create the benchmark against which the finished system will be evaluated. By clearly stating how the system is supposed to work, this chapter sets the stage for the testing procedures in Chapter 4, ensuring the final product meets its intended objectives.

3.1.2 Research design and approach

This study adopts a mixed-methods research design, incorporating both qualitative and quantitative approaches. Qualitative methods are utilized for understanding the contextual factors, user requirements, and design considerations, while quantitative methods are employed for evaluating system performance and user satisfaction. The approach is structured yet flexible, combining several methodological elements to ensure both academic rigor and practical utility.

The core of the work is the creation of a functional software artefact the web-based digital archive. This is classified as Applied Development Research, where the primary output is a novel technological solution (the system) designed to solve a concrete, real-world problem (the inefficient management of academic resources).

3.1.3 Methodology Used in the Proposed

The development of the web-based digital archive follows a **modified Waterfall model with iterative feedback loops**. This structured approach ensures systematic progression from requirements to deployment while allowing for refinements at each stage.

The methodology comprises five sequential but flexible phases:

1. Requirements Analysis & Planning
2. System Design
3. Implementation & Coding
4. Testing & Validation
5. Deployment & Documentation

Each phase produces specific deliverables that feed into the next, ensuring traceability and quality control throughout the development lifecycle.

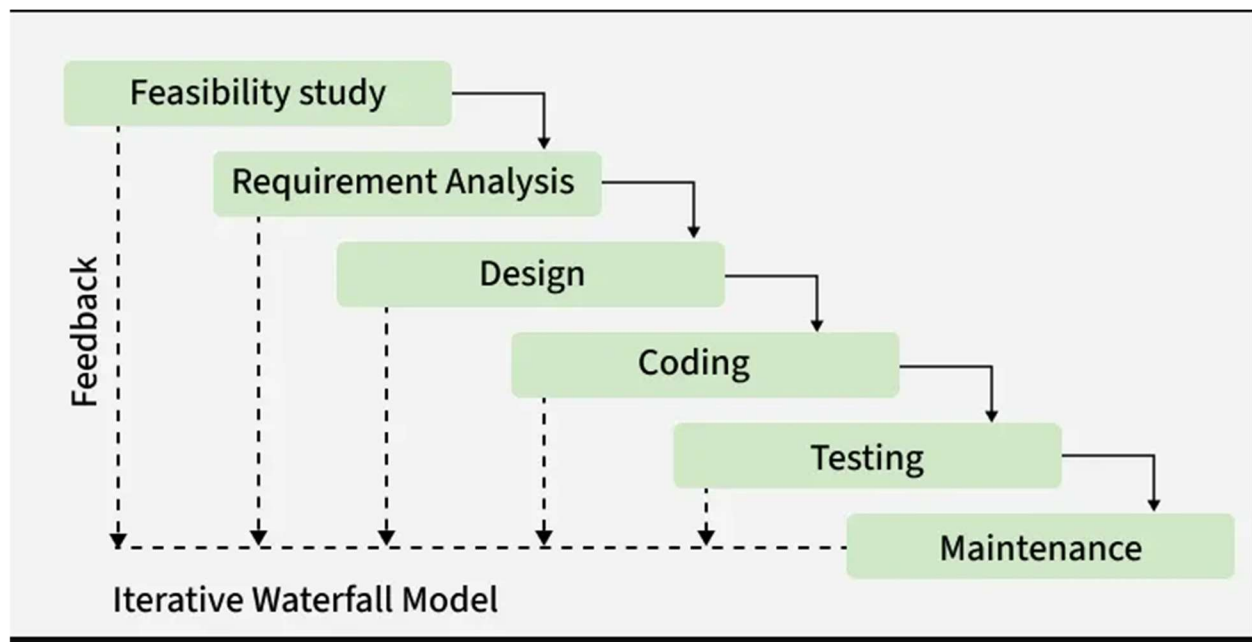


Fig 3.1 Modified Waterfall Methodology with Iterative Feedback Loops

1. Requirements Analysis

In this phase, the system requirements were gathered and analyzed. Problems associated with fragmented storage of past questions and learning materials were identified. Functional requirements (such as upload, search, filtering, and role-based access) and non-functional requirements (such as security, usability, and availability) were defined.

2. System Design

Based on the requirements, the system architecture was designed. This included database schema design, user interface structure, use case diagrams, and data categorization strategy. The Admin and Student dashboards were conceptualized during this phase.

3. Implementation

The system was developed using appropriate web technologies. The database was created, user authentication mechanisms were implemented, and the digital archive functionalities were coded.

4. Testing

The system underwent testing to ensure proper functionality. Errors identified during testing were corrected through feedback loops to the implementation or design phase where necessary.

5. Deployment and Evaluation

The completed system was deployed in a controlled environment for evaluation. Feedback was collected to assess usability, performance, and effectiveness in organizing academic materials.

3.2.0 Existing System

This section examines the current practices and methods used for managing and distributing past questions and learning materials at Edo State University, Iyamho, and similar institutions. Understanding these existing processes is crucial for identifying the specific problems that the proposed digital archive aims to solve.

The current system is characterized by informal, decentralized, and manual processes that rely heavily on interpersonal networks and basic digital tools not designed for academic resource management.

3.2.1 Analysis of Existing System

The current ecosystem for accessing past questions and learning materials at Edo State University, Iyamho, operates through a mix of formal and informal channels:

1. **Institutional Learning Management System (LMS):** The university's official platform (such as Moodle, Canvas, or a similar system) serves as the primary channel for current course materials. However, its use for past questions is typically limited and inconsistent.
2. **WhatsApp/Telegram Groups:** The most active method for sharing materials outside the LMS. Lecturers, class representatives, and students create subject-specific groups where

PDFs, slides, and past questions are shared as chat attachments. These groups operate parallel to the formal LMS.

3. **Personal Storage & Peer-to-Peer Sharing:** Students maintain personal archives on devices and share via Bluetooth, email, or flash drives. This creates fragmented, duplicated resources across the student population.

3.2.2 Disadvantages of Existing System

Resource Fragmentation: Students must navigate multiple platforms (LMS, WhatsApp, personal drives) to find complete study materials, wasting time and causing frustration.

Loss of Historical Materials: The LMS typically restricts access to course materials after semester completion. This means past questions and lecture slides from previous years become inaccessible, breaking the continuity of academic resources.

Lack of Standardization: Materials shared via informal channels have no consistent naming, formatting, or quality control. Students often receive multiple versions of the same document without knowing which is most accurate or recent.

Search Inefficiency: Neither the LMS (when configured for current courses only) nor WhatsApp groups offer effective search across years, departments, or resource types. Finding a specific past paper requires manual scrolling or asking peers.

Access Inequality: Students who miss WhatsApp shares, join groups late, or have limited data at the time of sharing are permanently disadvantaged. Similarly, departmental archives often have restricted access hours.

No Pre-Resumption Preparation: The LMS is inactive between semesters, and informal channels are quiet. This creates a complete blackout period where incoming students cannot access materials for their upcoming courses.

Security and Copyright Concerns: Open sharing on WhatsApp exposes lecturers' intellectual property without control. There is also no way to track or manage appropriate usage of materials.

Storage Limitations: WhatsApp compresses files and may automatically delete older media. Personal device storage is finite and vulnerable to loss or damage.

No Central Administration: There is no systematic way for the university to update, verify, or retire materials. The quality and relevance of resources depend entirely on individual initiative.

3.2.3 Structure of Existing System

Figure 3.1 illustrates the schematic diagram of the existing system. The system consists of tracking of learning, offline course materials, ...

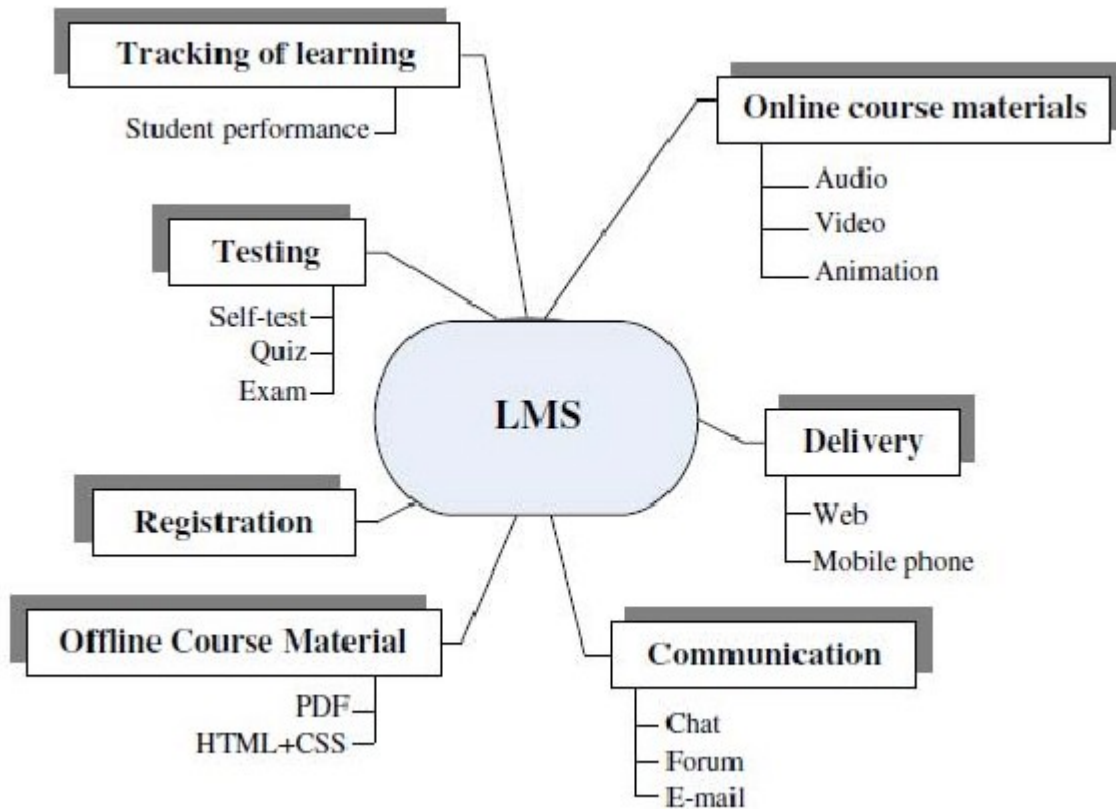


Figure 3.2: Existing system

The existing system is primarily based on the institutional Learning Management System (LMS). The structure of this system revolves around course-based access rather than centralized archival storage.

In the LMS structure, students log into the platform and access materials through individual course pages. Each course is managed independently by the assigned lecturer, who uploads lecture notes, assignments, and occasionally past questions. Materials are therefore organized per semester and per course.

The system structure follows a hierarchical model:

- Administrator – Manages overall platform configuration and user accounts.
- Lecturers – Upload course materials within their assigned course spaces.
- Students – Access materials only for courses they are registered for in a particular semester.

One key structural characteristic of the LMS is that materials are session-based. Once a semester ends, access to certain materials may be restricted, archived, or removed depending on system settings. This means resources are not persistently organized across academic years.

3.3.0 Proposed System Design

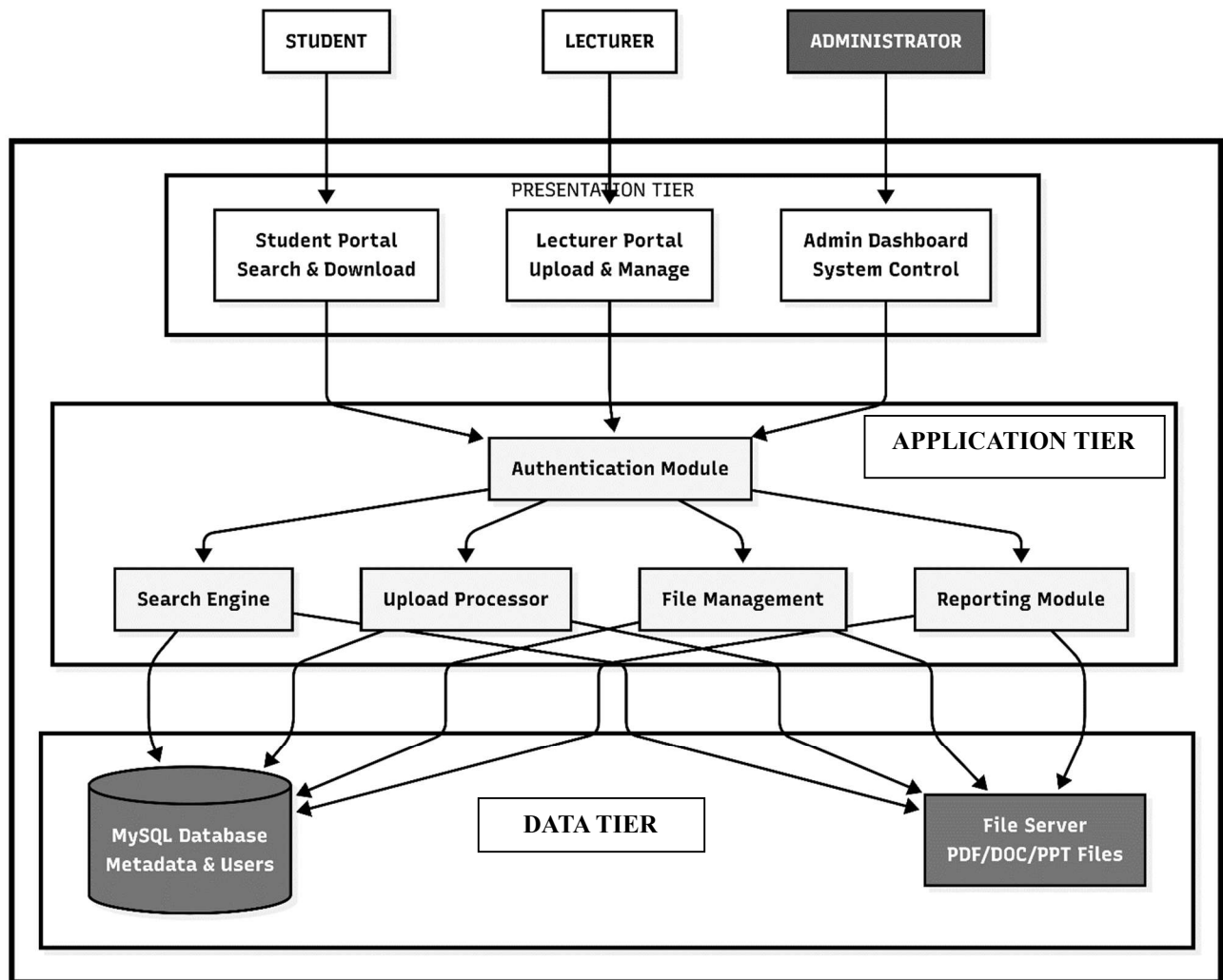


Figure 3.3: Proposed system model

Figure 3.3 illustrates the three-tier architecture of the proposed web-based digital archive. The Presentation Tier provides role-specific interfaces for students, lecturers, and administrators. The Application Tier handles all business logic including authentication, search, and file processing. The Data Tier manages both structured metadata in a MySQL database and unstructured file storage for academic materials.

3.3.1 Analysis of Proposed System

The system employs a modular architecture with the following core components:

1. **User Management Module:** Handles registration, authentication, and role-based access control for students, lecturers, and administrators.
2. **Content Management Module:** Manages the upload, categorization, and organization of past questions and learning materials.
3. **Search and Retrieval Module:** Provides advanced filtering and search capabilities across multiple parameters.
4. **Database Management Module:** Ensures efficient storage, indexing, and retrieval of both metadata and actual files.
5. **Reporting Module:** Generates usage statistics and system performance reports.

3.3.2 Advantages of Proposed System

First, it offers **centralized access**, bringing all past questions and learning materials together on a single platform. This removes the stress of searching through multiple channels such as learning management systems, WhatsApp groups, or personal storage devices. The archive also supports **persistent storage**, meaning materials remain available across academic sessions and years. This addresses the common problem of losing access to resources once a semester ends or an LMS course is closed. With its **efficient search functionality**, students can quickly locate materials using filters such as course code, department, level, semester, year, or keywords. What previously took several minutes or even hours can now be achieved within seconds.

In addition, the system promotes **standardized organization** through consistent naming conventions, proper categorization, and metadata tagging. This ensures that materials are easy to understand and uniformly presented. The inclusion of **version control** helps eliminate confusion caused by multiple copies of the same material. Students can easily identify the most recent and accurate version of any resource.

Another key advantage is **pre-resumption access**, which allows students to prepare for upcoming courses before the semester begins, improving academic readiness and confidence. The system also

reduces the risk of data loss through **institutional backup mechanisms**, protecting materials from accidental deletion, device failure, or file expiration.

Through **controlled access**, role-based permissions ensure that intellectual property is protected while still allowing appropriate access for different users. The web-based design makes the system **scalable**, allowing it to grow alongside the number of users and materials without requiring major infrastructure changes.

Finally, **usage analytics** enable administrators to monitor frequently accessed materials, identify resource gaps, and make informed decisions about content management.

3.3.3 Structure of Proposed System

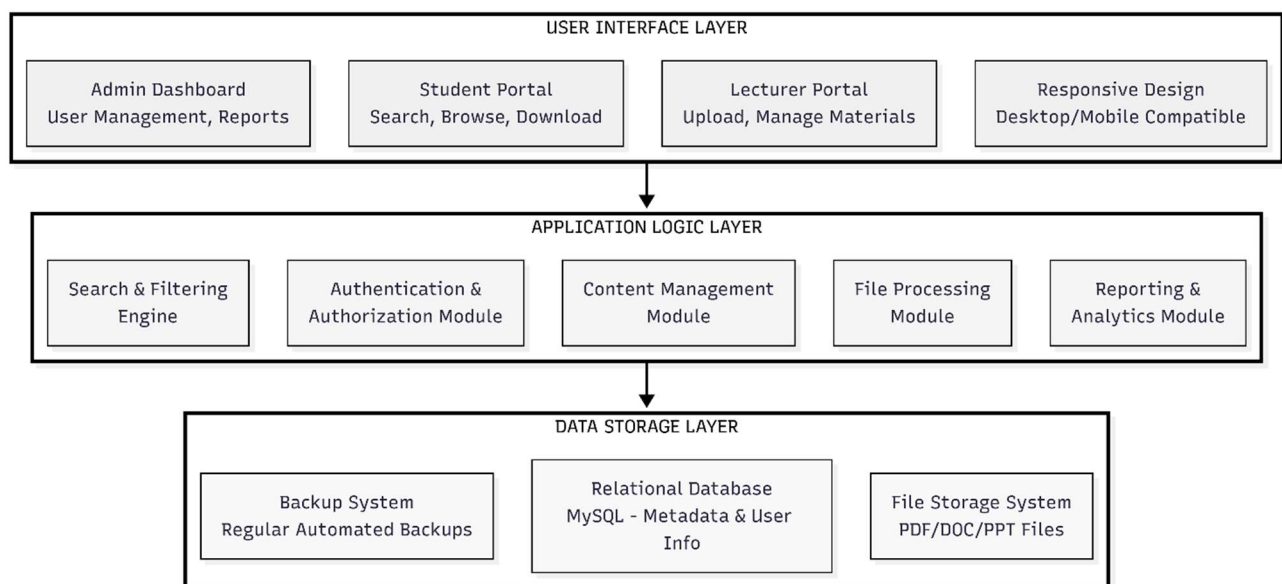


Figure 3.4

3.4.0 Data Analysis Methods

The system employs both **structured data analysis** (for metadata and user information) and **file content management** (for the actual academic materials). Data flows through several key processes:

1. **Data Categorization:** All uploaded materials are automatically tagged with metadata including
 - Course code and title
 - Academic department
 - Level (100-500)
 - Semester (1st or 2nd)
 - Academic year
 - Material type (past question, lecture slide, note, etc.)
 - File format (PDF, DOC, PPT, etc.)
2. **Data Validation:** Each upload undergoes validation checks:
 - File format verification
 - File size limits (to prevent server overload)
 - Duplicate detection (checking for existing identical materials)
 - Metadata completeness (ensuring all required fields are filled)
3. **Search Optimization:** Data is indexed for efficient retrieval:
 - Full-text indexing of PDF and document contents where possible
 - Keyword extraction from filenames and metadata
 - Creation of search indices for fast query responses
4. **Usage Analytics:** The system collects and analyzes usage data:
 - Download frequency per material
 - Popular search terms
 - User access patterns
 - Peak usage times

3.4.1 Description of data analysis techniques

The digital archive system uses straightforward data analysis approaches to keep everything organized and useful.

When a file gets uploaded, the system pulls out important details like the course code, academic year, department, and what type of material it is. These details become the backbone of how students search and filter through everything in the archive. It's what makes finding a specific past paper from 2022 so much easier than scrolling through endless WhatsApp messages.

For documents that contain actual text, like PDFs and Word files, the system creates a searchable index of the words inside. This means you're not just searching filenames; you can actually search for keywords that might be mentioned in the document itself. It's not super complex text mining, but it gets the job done and makes the archive much smarter.

The system quietly keeps track of how people use it. Which materials get downloaded the most? Are students searching more right before exams? Which departments have the busiest archives? This isn't about tracking individual students—it's about understanding overall patterns so the system can be improved and expanded where it's needed most.

To keep everything running smoothly, the system runs regular checks in the background. It makes sure files haven't gone missing, flags potential duplicates before they clutter up the archive, and ensures all the necessary information is filled in properly. Think of it as a quality control system for academic resources.

Simple statistics help turn raw data into useful information. How many materials does the Computer Science department have compared to Mathematics? How quickly is the archive growing? How many people are actually using it? These basic numbers help everyone understand the health and value of the archive.

The system can even spot basic patterns over time. It notices when exam season is approaching because downloads spike. It can point out which courses might need more materials added. It shows which types of resources are most popular. These aren't complex predictions, but they're practical insights that help keep the archive relevant and useful.

Together, these techniques ensure the archive isn't just a digital filing cabinet—it's an organized, searchable, and constantly improving resource that adapts to what students and lecturers actually need.

3.4.2 Method of data coding and categorization

Data coding and categorization in the web-based digital archive were carried out to ensure that past questions and learning materials are well-organized, easy to retrieve, and consistently stored. Each uploaded material was assigned a structured set of attributes at the point of entry into the system.

For every academic resource, key metadata such as course code, course title, department, academic level, semester, academic year, and material type were captured. These attributes served as unique identifiers that allowed the system to classify and group materials accurately within the database.

To maintain uniformity, standardized naming conventions were adopted during the coding process. Files were named using a consistent format that combined relevant attributes such as course code, academic year, and material category. This approach reduced ambiguity and prevented duplication of records within the archive.

Categorization was implemented using predefined database fields and relational tables. Past questions and learning materials were grouped according to their respective departments and courses, while additional filters such as level and semester further refined classification. This structured categorization supports efficient querying and fast retrieval of materials through the system's search functionality.

The coding process also enabled basic version control by allowing newly uploaded materials to be linked to existing records, ensuring that the most recent and relevant versions were accessible to users. Access permissions were applied based on user roles, ensuring that only authorized users could upload, modify, or delete archived materials.

Overall, the method of data coding and categorization adopted in this system improves data consistency, enhances search efficiency, and supports the long-term integrity of the digital archive.

3.5.0 User Design Phase

The user design focused on creating an intuitive interface that addresses the needs of all user groups. Through informal requirement gathering with students and lecturers, key pain points were identified: difficulty finding materials, cluttered interfaces, and complex upload processes.

The design emphasizes:

- **Simplicity:** Clean layouts with minimal clutter
- **Accessibility:** Clear navigation and readable text sizes
- **Responsiveness:** Works well on both desktop and mobile devices
- **Role-based workflows:** Different interfaces for students, lecturers, and admins

Wireframes were created for key pages: homepage, search results, upload form, and admin dashboard, ensuring the system would be easy to use even for non-technical users.

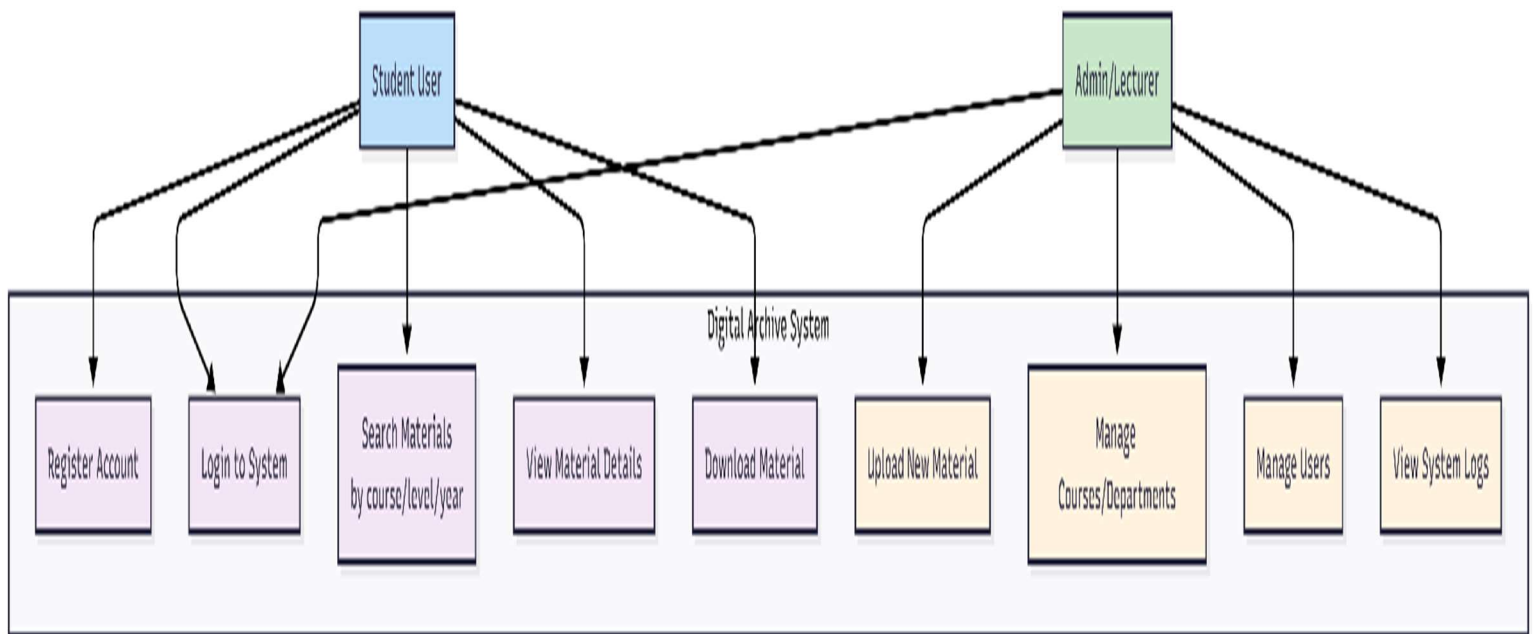
3.6.0 Use Case Model

Student Actors Can:

- Register and log in to their accounts
- Search and browse archived materials
- Filter results by course, department, level, or year
- View material details and download files
- Update their profile information

Lecturer/Admin Actors Can:

- Upload new past questions and learning materials
- Categorize uploaded files with proper metadata
- Manage existing materials (edit, update, or remove)
- View download statistics and usage reports
- Manage user accounts and permissions



3.8.0 Algorithms

1. User Authentication Algorithm

BEGIN Authentication (username, password)

1. Query database for user with matching username

2. IF user not found THEN

RETURN "Invalid credentials"

3. ELSE

Retrieve stored hashed password

IF hash(password) == stored hash THEN

Generate session token

Store session in database

RETURN success + user role + token

ELSE

RETURN "Invalid credentials"

4. END IF

END Authentication

2. File Upload and Categorization Algorithm

BEGIN UploadFile(file, metadata, uploader_id)

1. Validate file type (allow: PDF, DOC, PPT)

2. Check file size (< 10MB)

3. Extract metadata: course_code, year, level, type

4. Generate unique filename: course_year_type_timestamp.ext

5. Save file to server storage

6. Insert record into database:

- File path

- Metadata

- Uploader ID

- Upload timestamp

7. Update search index with new file

8. RETURN upload_success + file_id

END UploadFile

3. Search and Retrieval Algorithm

BEGIN SearchMaterials(filters, keywords)

1. Parse search parameters:

- Department (optional)

- Course (optional)

- Level (optional)
- Year (optional)
- Keywords (optional)

2. Construct database query with filters
3. IF keywords provided THEN
 - Search indexed content for keyword matches
 - Combine results with filtered results
4. Sort results by relevance:
 - Exact course matches first
 - Recent materials prioritized
 - Popular downloads weighted higher
5. Paginate results (20 items per page)
6. RETURN formatted results

END SearchMaterials

4. Access Control Algorithm

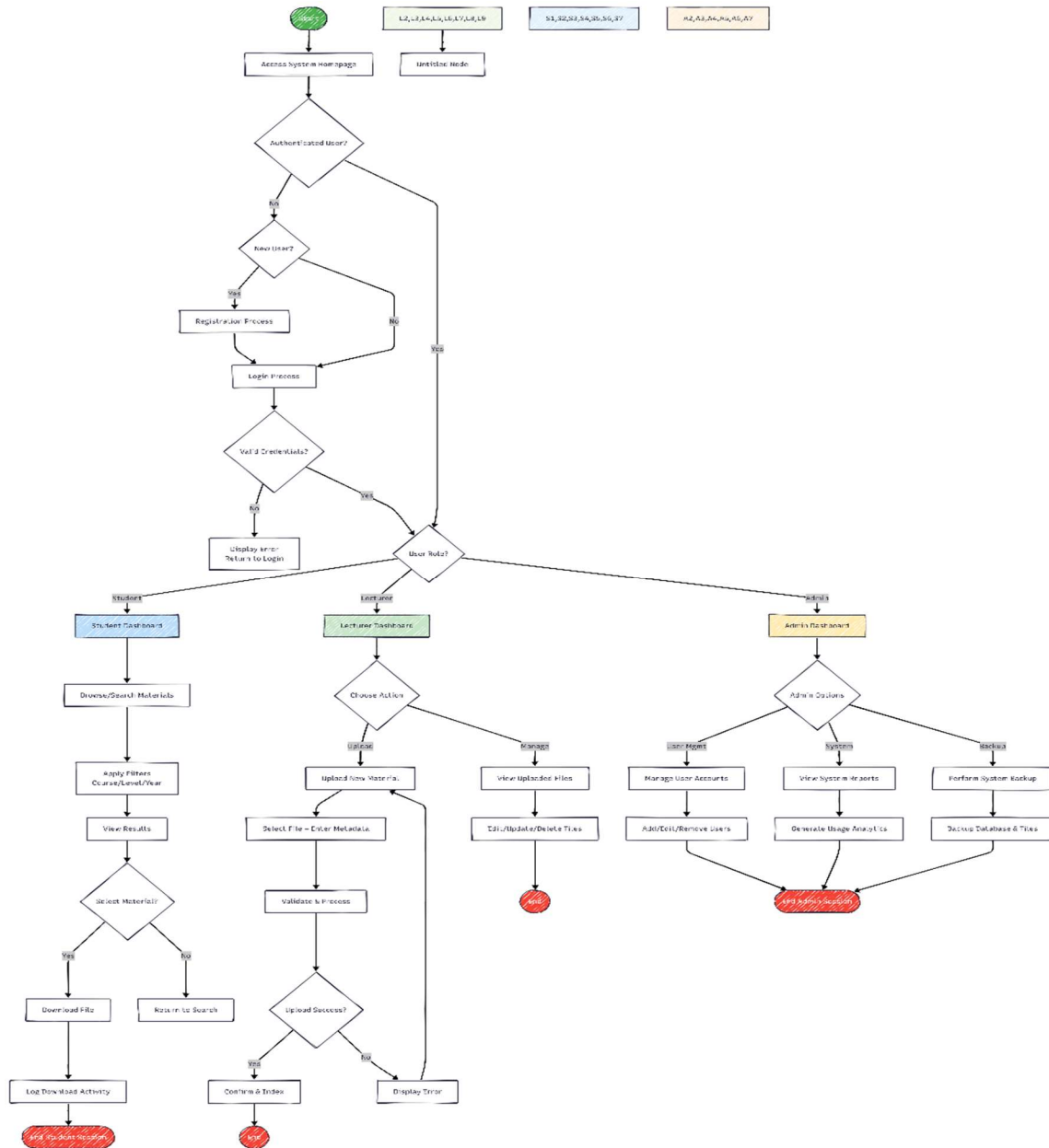
```
BEGIN CheckPermission(user_id, resource, action)
1. Get user role from session
2. SWITCH (user_role)
  CASE "student":
    IF action == "download" AND resource is public THEN
      RETURN true
    ELSE
      RETURN false
  CASE "lecturer":
    IF action == "upload" OR
      (action == "edit" AND user owns resource) THEN
      RETURN true
    ELSE
      RETURN false
  CASE "admin":
    RETURN true // Full access
3. END SWITCH
END CheckPermission
```

5. Duplicate Detection Algorithm

```
BEGIN DetectDuplicate(file_hash, course_code, year)
1. Calculate file hash (MD5/SHA-256)
2. Query database for existing files with:
  - Same file hash OR
  - Same course_code + year + similar filename
3. IF duplicates found THEN
  RETURN list_of_duplicates + suggestion
4. ELSE
  RETURN no_duplicates
END DetectDuplicate
```

3.9.0 Logic Design (Flowchart)

The flow chart for the proposed system starts from the admin section who is responsible for managing all students, managing exam officer, departmental lecturers, courses, upload exam questions.



3.10 Chapter summary

This chapter has outlined the comprehensive design and methodology for developing the web-based digital archive. From analysing the fragmented existing system to proposing a centralized solution, the design addresses all identified gaps through a structured, three-tier architecture. The system's modular components, clear algorithms, and intuitive user workflows create a solid foundation for implementation. The key methodological decisions about the proposed system are as follow:

1. **Architecture Choice:** Selected a three-tier web-based architecture for scalability and accessibility.
2. **Development Approach:** Adopted an iterative design process with continuous feedback integration.
3. **Technology Stack:** Chose LAMP/MEAN stack components for reliability and community support.
4. **Database Design:** Implemented a relational database structure with proper normalization.
5. **User-Centric Design:** Prioritized simplicity and responsiveness across all interfaces.
6. **Security Model:** Incorporated role-based access control from the ground up.

In addition, the implications for study's validity and reliability are as follows:

Validity Implications:

- The design directly addresses the research objectives stated in Chapter 1.
- Each system component corresponds to identified user needs from the problem analysis.
- The methodology ensures the final product will solve the real-world problem it was designed for.

Reliability Implications:

- Standardized coding practices ensure consistent system performance.
- Modular design allows for independent testing of each component.
- Clear documentation supports future maintenance and upgrades.
- The structured approach enables replication by other institutions.

Essentially, the design presented in this chapter provides a technically sound, pedagogically appropriate, and practically feasible solution to the academic resource management challenges identified earlier. It balances innovation with practicality, ensuring the system will be both effective and sustainable.